Development since the previous PDFs

26 September 2026

Completed code and finite computational evidence; active Phase 4 simulations; concrete design contracts for Phase 5.

How to read this report

The narrative comes first. Appendices review the remaining plans and specify the next intervention contracts and task map. Proposed design, verified implementation and sampled performance retain different evidence statuses.

Latest PDF baseline: novelty register, 24 September. Last broad technical snapshot: completed Phase 2, 23 September. Both historical PDFs are preserved.

Fixed input inventory (SHA-256)

1. docs/development-update-20260926.md
`98ba09d2b0a4a74ace87688a17a7bee94e2156ad5cd0039db7aa32f756456068`

2. docs/phase4-plan-review-and-next-design.md
`9b64660f85d63ce0ae851a93a79e745f653699f53ec3403bc920750b504e4992`

3. .planning/phases/05-guided-and-perturbative-bp-interventions/05-DESIGN.md
`de687c5b1fd9d88307e7aca88376ff9f383e6ace1652ad920dd21305d517db49`

4. .planning/phases/05-guided-and-perturbative-bp-interventions/05-DESIGN-TASKS.md
`427c223a6746c062d483ff0d3e99926d3be1d8df7ccfeac58a2036b9741408cc`

5. docs/evidence/development-snapshot-20260926.json
`b09828dce050a29f4be653ebfa43ca9221e06543eb5de0d5ff2f050145461e51`

6. docs/evidence/phase4-low-noise-design-snapshot-20260926.json
`4599e809e12da24aee9127c2d22e6e89462c8d76fc3b74fc099aae155ee03f0a`

7. docs/evidence/phase5-source-status-20260926.json
`601892400d6324e8609b7a4a7ba96c98737a410b18312e5cb9164fbb5f524dc9`

8. docs/evidence/intervention-design-validation-20260926.json
`9234dd3bbf0607bf41a3db2912eacaebcb7a74472b136f31c07e9e14e970add8`

The Markdown files are editable sources. JSON inputs freeze observed status, source availability and algebra checks. Rebuilding does not read changing calibration files.

Development and computational evidence update

Policy-Based BP Research Platform for QLDPC Decoding

26 September 2026. Evidence snapshot: 22:47 UTC. Active work: Phase 4, Plan 12. This report includes completed implementation work, pending simulations and preparatory Phase 5 design, with those statuses explicit.

1. What changed since the previous PDFs

The latest dated PDF is docs/research-novelty-and-depth-register.pdf, recorded on 24 September in commit 2a83502. It added a prospective quaternary memory research question. The latest broad technical PDF, docs/policy-bp-documentation-through-phase-2.pdf, was recorded on 23 September in 30df961 and ended at Phase 2. This update bridges that broader snapshot with a short Phase 3 account and then documents the Phase 4 developments since the novelty PDF. Both historical PDFs retain their exact bytes.

The main changes are a verified update/scheduling engine, active-memory Relay execution and native-reference diagnostics, bounded calibration and lower-noise estimation, a discovered ordinary-memory dispatch bug and its isolated correction, guarded fresh-evaluation tooling, and an explicit design for the next intervention phase.

The repository snapshot used here is main commit af09c5e. The memory correction is commit fc214ef in a separate worktree; it has not been merged into the active simulation source. Frozen data, source paths, original design bytes, historical PDF hashes and capture time are retained in docs/evidence/development-snapshot-20260926.json.

2. Phase 3: update rules and scheduling

Phase 3 completed on 24 September with all 17 plan summaries and its verification gate passed. It added min-sum, normalized min-sum and offset min-sum conventions; layered, check-serial and variable-serial schedules; residual priorities and cache semantics; randomized ordering with replayable streams; and a source-qualified Hsu group schedule. These remain policies inside the compatible canonical binary engine.

The phase verification recorded 300 passing tests. Retained policy pilots contain 19 verified one-shot terminals, plus two Hsu terminals. Their value is checking equations, ordering, compatibility, rollback and provenance. Those tiny pilots are not statistical curve reproduction. Hsu's schedule is a declared binary CSS adaptation of its classical problem. The separate Moradi qualifier remains unavailable without the author partition and Q-table.

Authority: .planning/phases/03-update-rules-and-scheduling/03-VERIFICATION.md. These are historical Phase 3 gate counts, not a new full-suite run for this PDF.

3. Three comparisons that must remain distinct

Arm	Execution	Meaning
Source-procedure	Canonical Python engine, min-sum	Declared realization of the source's BP procedure with CSS adaptation limits
Canonical-default	Canonical Python engine, sum-product	Same engine family, different update rule and possibly different stopping/work

Arm	Execution	Meaning
Author-native	Separately staged author Rust decoder	Executable reference pinned to relay v0.2.2

The native author pin is 14cb31a18f9b9d7d099791b793e35b8a2951f8f3. Native environments remain isolated from canonical execution. Source-procedure does not mean that the GitHub Rust code ran; it means that the specified source behavior was expressed in our canonical engine. Native evidence is named explicitly when that code actually runs.

The controlled source-procedure versus canonical-default comparison changes the BP update rule from min-sum to sum-product while preserving common declared controls. Both run in Python. Their timing differences can include update cost, convergence paths and amount of work. A separately tuned comparison may additionally use different selected coefficients; it must disclose that.

Author-native versus canonical comparisons have a Rust/Python implementation difference and can also differ in numerical conventions, trajectories, instrumentation and timing scope. Recorded pipeline ratios cannot isolate a language effect.

4. Memory and Relay development

Relay executes affine-prior memory dynamically at source-iteration boundaries, with uniform first-leg and declared disordered later-leg parameters. Sequential legs transfer the previous posterior; new leg messages and private histories follow explicit reset rules. Source and canonical numerical profiles, gamma identities, leg work and terminal reasons remain visible.

Artifact qualification resolved the B1 identity discrepancy instead of treating matching code parameters as identical matrix evidence. The benchmark/caps work now uses reconciled B1, BB-144 and generated HGP artifacts with their different source-status qualifications. Bounded calibration selects settings using retained outcomes and work under a frozen rule. Fresh evaluation has its own sample namespace.

Ordinary-memory execution correction

During continued coding, the public ordinary-memory V4 runner was found to validate memory bindings and then discard them in its compact V3 projection. Such runs executed plain BP. The isolated correction uses full compact V5 dispatch and executes affine-prior, V2C damping, PDBP, PD-prime and momentum memory at whole flooding-sweep boundaries.

The correction adds private per-binding histories, atomic budget/state commit, explicit prior -> posterior/belief -> V2C operand ordering, history digests and element-work accounting, and independent semantic reconstruction. Momentum now exposes the required alpha/gamma parameters. PD-prime uses an O(n) sufficient statistic with declared first-field/prior assumptions.

Failed budget/iteration-limited runs retain the last committed diagnostic candidate and its origin; it remains unselected when invalid. Rehashed semantic tampering is rejected. Numerical failure and denied work preserve the appropriate committed prefix and private histories.

The correction handoff records a regression run of 356 passed, one existing skip, then a final focused memory/manifest gate of 67 passed. These are implementation fixtures on fc214ef, not newly simulated campaign shots. The existing skip concerns the unfinished Phase 4 notebook identity workflow.

The live Relay campaign uses a separate dispatch and was unaffected by this ordinary-memory bug. Historical ordinary-memory results retain their original plain-BP meaning. Additional local schedules, history

lifetimes, retained histories, ordinary disordered assignment and per-iteration NMS still need execution support; fixture success does not close those gaps.

5. Computational evidence available now

Completed lower-noise estimation

This is the BB-144 independent X-only CSS code-capacity experiment, with four arms receiving the same identified physical samples at each rate and shot. It uses fixed, uncalibrated settings: first gamma 0.125, later center 0.25, width 1.25, an 80-iteration first leg, 60-iteration later legs, up to 31 Relay legs and one solution. Baseline arms have one leg. The numerical profiles are float64 SPA or min-sum as named in Section 3.

Physical error rate	Arm	Shots	All-shot failures	Failure LER	Exact interval, percent
0.05	SPA baseline	8192	337	4.114%	3.512–4.782
0.05	SPA Relay	8192	289	3.528%	2.971–4.152
0.05	Min-sum baseline	8192	353	4.309%	3.693–4.991
0.05	Min-sum Relay	8192	290	3.540%	2.982–4.165
0.06	SPA baseline	2048	198	9.668%	7.889–11.679
0.06	SPA Relay	2048	178	8.691%	7.002–10.619
0.06	Min-sum baseline	2048	219	10.693%	8.826–12.787
0.06	Min-sum Relay	2048	186	9.082%	7.356–11.044

The retained design prespecified six looks: 256, 512, 1024, 2048, 4096 and 8192 shots; at least 50 failures and relative interval half-width at most 0.35 for both all-shot failure and valid-selected conditional logical error in all four arms at a rate. Its two-sided Clopper–Pearson intervals spend alpha across looks and the two precision metrics at each arm/rate. Cell intervals have their separate correction. The table uses the stored primary intervals; it does not claim simultaneous coverage across every arm and rate.

The p=0.06 cohort reached these precision criteria at 2048 shots per arm; p=0.05 reached them at 8192. Reconciliation finished at 19:07 UTC on 26 September with status completed_precision. There was no prespecified cross-arm difference/ranking test. Smaller point estimates do not establish statistically significant improvement. This is an uncalibrated controlled CSS adaptation, not source-curve reproduction or a universal decoder ranking.

Result authority: data/results/phase4-memory-relay/low-noise-ler-reconciled-20260926/result.json. The original design lives in the separate low-noise worktree at c94064e; a hash-verified byte-identical copy is packaged as docs/evidence/phase4-low-noise-design-snapshot-20260926.json.

Outcome categories and denominators

The reported diagnostic proposal categories are:

- Valid-correct: original syndrome matches and logical action agrees.

- Valid-wrong logical: syndrome matches but logical action disagrees.

- Invalid-correct projection: syndrome fails while the projection onto the pinned logical basis agrees. This is not a valid recovery.

- Invalid-wrong projection: syndrome fails and that projection disagrees.

- No proposal: no retained diagnostic bit vector exists to classify.

Invalid candidates are not assigned code-space logical cosets. Their logical projection is a separate, basis-dependent diagnostic. A diagnostic proposal and a selected recovery are also different objects: BP can propose a bit vector that selection rejects because its syndrome is invalid. That is not the same as producing no vector.

At p=0.05 the five cells are:

Arm	Valid-correct	Valid-wrong	Invalid-correct	Invalid-wrong	No proposal
SPA baseline	7855	155	0	182	0
SPA Relay	7903	289	0	0	0
Min-sum baseline	7839	118	2	233	0
Min-sum Relay	7902	290	0	0	0

All rows sum to 8192. All-shot failure counts every shot lacking a selected, syndrome-valid, logically correct recovery, including no recovery/no proposal. Conditional logical error uses only selected syndrome-valid recoveries. For example, min-sum baseline has 353/8192 all-shot failures but 118/7957 conditional logical errors. Its 235 invalid diagnostic proposals are not zero-error recoveries. Both denominators and all cells must remain visible.

Native-author diagnostics are present

A completed controlled diagnostic has 128 matched canonical/native pairs for each of the three eligible codes. Native failures/timeouts were zero in these retained pairs. The source pin and environment origin are recorded; a separate bounded author test timed out and remains inconclusive rather than passing.

Code	Matched pairs	Success disagreements	Exact candidate disagreements
Reconciled B1	128	10	37
BB-144	128	3	16
Generated HGP	128	9	65

Different bit vectors can still be valid equivalent recoveries. Conversely, similar aggregate success rates do not establish identical procedures. The retained diagnostics quantify disagreement and preserve uncertainty; 128 pairs are not an equivalence verdict or a source circuit-curve reproduction. The result is in author-v022-agreement-v1/result.json, frozen in the snapshot.

Calibration remains pending

At 22:47 UTC the calibration process was active after approximately 25 h 48 m. Its serialized result retained 1750 attempt records: 672 in each B1 preset and 406 in the BB-144 source-procedure preset. The plan contains 182 cells; the result had 55 configuration records, including an unfinished cell. These are snapshot

counts, not a finalized completion percentage. The evaluation namespace had not been accessed and no terminal selection was claimed.

The long duration reflects multiple parameter configurations, matched shots, possibly many BP/Relay units per shot and mandatory verification. The Python implementation and process/setup boundaries also contribute. Trying several gammas is one source of work; the recorded evidence does not isolate it as the sole runtime cause. Increasing workers helped the separate lower-noise campaign preserve its design while running four arms concurrently.

6. Runtime evidence and the next measurement

The completed native comparison already contains pipeline timings for matched observations. Its median per-pair canonical/native ratios are:

Code	Median recorded ratio	Canonical mean, seconds	Native mean, seconds
Reconciled B1	1.804	64.733	36.894
BB-144	1.661	6.243	2.464
Generated HGP	2.065	5.594	2.751

The canonical timer covers the public CLI run plus mandatory verification. The native timer covers observation preparation, gamma reconstruction, the isolated author process/decoder and diagnostic classification. Calls were canonical-first. Thus the native recorded pipeline was faster on these samples, but these unequal boundaries do not give decoder-only speed or the cost attributable to language.

The guarded fresh-evaluation runner in 43dafb5 includes separately timed native decode calls and separate preparation/verification facts. An equal-boundary comparison still needs canonical decoder-only calls timed at the same input/output boundary with matched algorithm, controls, numerical profile, stopping and thread/resource settings. Counterbalance call order, disclose warmup, and report candidate outcomes and work alongside latency. Missing native internal counters remain missing, not zero.

Authority: author-v022-agreement-v1/runtime-comparison.json; the snapshot preserves its exact scope declarations and summary digest.

7. Plan review and forward design delivered here

The review identifies the unfinished calibration/evaluation/notebook gates, the ordinary-memory integration prerequisite, factorial ordering discrepancy, shared-view admission requirement and missing trace support. Plan 04-14's coverage table describes intended coverage, not completed evidence. Details are in the plan-review appendix.

Phase 5 now has proposed contracts for interventions, transformed working problems, fixation, BP continuation, certified lifts, original validation, restart streams, atomic commit, lineage and separate work categories. Its next-task map covers MOD-02, MOD-03 and INT-01–04. Primary-source ambiguities, native availability and unresolved flip variants remain explicit.

The structural shortening proposal has an executable independent algebra check: 16,384 exhaustive finite cases passed, together with four constructed cases covering a misleading working syndrome, a certified lift, all-fixed success/contradiction and finite-prior behavior. The retained fixture report is docs/evidence/intervention-design-validation-20260926.json. These checks validate a design equation, not a

new decoder implementation or LER claim.

The key extension is that current WorkingProblem accepts only identity graphs/syndromes. Interventions need a versioned working-view and lift API; candidate validation must always reach the immutable original equation. The new task map makes that dependency concrete before BPGD or impulse coding.

8. Next execution sequence

1. Continue the isolated Phase 4 ablation/trace implementation while the live calibration retains its source identity.

2. Preserve original factorial design/caps and freeze an outcome-independent execution supplement for coefficients and operand order before sampling.

3. Reconcile terminal calibration, freeze selections and admit any extra views explicitly; execute fresh matched evaluation and native reference pairs.

4. Complete ablations, bounded traces, the executable notebook and Phase 4 verification gates, then integrate the isolated correction safely.

5. Convert the Phase 5 design into checked execution plans after Phase 4. Source discovery and interface design can continue meanwhile.

Beam/search remains Phase 6, optional fallback Phase 7, campaign qualification Phases 8–9, correlated quaternary Phases 10–11, circuit decoding Phases 12–13 and publication Phase 14. All three regimes remain in the journal scope.

9. Rebuilding and interpreting this report

Run .venv/bin/python tools/build_development_pdf.py to rebuild this dated PDF, or add --check to compare exact generated bytes. Its fixed Markdown inventory includes this narrative, the plan review, Phase 5 design and task map; a source manifest prints their hashes plus the retained evidence hashes. PDF generation reads the frozen documentation snapshot rather than live simulation progress, so later results cannot silently alter this report.

Read the detailed design's primary-source links for mathematical definitions. Repository verifiers and retained ledgers remain the evidence authority. Source interpretations and proposed contracts are explicitly separated from implemented behavior and computational measurements.

Plan review and next design work

Snapshot: 26 September 2026. Active execution remains Phase 4, Plan 12.

What the plans require

The roadmap records Phases 1–3 complete and 12 of 15 Phase 4 plan summaries. The remaining work is correctly sequenced: terminal calibration and selection (04-12), fresh evaluation plus memory/Relay ablations and traces (04-13), then the executable notebook and public research guide (04-14). Completed low-noise estimation and native diagnostics are separate evidence; they do not close these three plans.

Plan or prerequisite	Current evidence	Remaining gate
04-12 calibration	Running, evaluation namespace untouched	Terminal reconstruction and frozen selections
Ordinary-memory execution	Corrected in isolated commit fc214ef	Integrate after preserving campaign source identity
04-13 selected evaluation	Guarded runner exists, commit 43dafb5	Terminal selection, admitted views and fresh sampling
04-13 factorial	Frozen design; corrected flooding primitives	Versioned order supplement, runner, missing bindings and traces
04-14 notebook	Planned sections and identity rules	Implement and execute against completed upstream evidence
Phase 5	Preparatory contracts and algebra checks in this update	Source resolution and Phase 4 completion before phase execution

Findings that change the next steps

A validated memory binding did not guarantee executed memory. The main V4 ordinary-memory path projected to V3 and dropped memory bindings. The isolated correction uses the full compact V5 dispatch. Older ordinary-memory ledgers remain plain-BP executions and cannot demonstrate a memory effect. The separate Relay path is unaffected. See .planning/phases/04-memory-policies-and-relay-bp/04-CODING-RESUME-HANDOFF.md.

The correction has a defined scope. It executes all five ordinary memory families at flooding boundaries, commits histories atomically, retains diagnostic candidates and verifies their semantics. It does not complete local memory schedules, additional history lifetimes, retained history across legs, the ordinary disordered-affine assignment path or per-iteration NMS. Scalar/graph NMS and per-iteration OMS are covered. An unsupported binding must be rejected explicitly rather than silently simplified.

The factorial order needs an outcome-independent supplement. The frozen interaction arrays put damping or momentum before affine-prior memory. The executable operand dependency is prior -> belief/posterior -> V2C. Preserve the frozen design and caps; add a versioned execution interpretation, explicit coefficients and stable binding order before sampling. The design cannot be changed in response to favorable outcomes.

Shared settings are not automatically an admitted evaluation arm. The runner guards terminal calibration, cap hashes and source pins, but an additional shared-settings view needs an explicit caps/design

amendment. Report per-preset reproduction-oriented views separately from controlled settings. Native comparisons must name the Rust reference and its translation.

Plan coverage is a promise of future evidence. The coverage table in 04-14 describes its intended implementation. It is not an execution audit. The skipped notebook identity test in test_phase4_evidence.py is still a placeholder; its old B1-digest explanation is stale after reconciliation. Replace it with executable stage-reuse checks when the notebook exists.

Ledger transformation slots are not an executable transformed problem. canonical/lineage.py already names transformations and lifts, but canonical/problems.py accepts only identity WorkingProblem graphs and syndromes. Phase 5 needs a typed transformation contract and independent original-problem validation before interventions can run.

Work that can proceed while results are pending

The following work depends on contracts and sources, not measured LER:

1. Define immutable working views, fixation maps and certified candidate lifts.

2. Specify BP round continuation, prior-vector changes and cache invalidation.

3. Resolve source variants and native-code availability for BPGD and impulse.

4. Specify rollback, restart streams, work reservation and original validation.

5. Prepare deterministic algebra fixtures and future implementation acceptance checks. The exhaustive shortening checks delivered here are design evidence.

6. Design equal-boundary runtime collection and honest missing instrumentation.

The detailed proposal is .planning/phases/05-guided-and-perturbative-bp-interventions/05-DESIGN.md; bounded next tasks are in 05-DESIGN-TASKS.md. These documents do not mark Phase 5 implemented or relax the Phase 4 dependency.

Execution order after this update

Keep the active campaign's code identity stable. Continue Phase 4 coding in the isolated correction worktree, finish the ablation/trace prerequisites, then reconcile terminal calibration and freeze settings. Run admitted fresh evaluation and native pairs, complete the notebook and verification gates, and then turn the Phase 5 design into checked execution plans. Preserve failed, incomplete and inconclusive evidence throughout.

Quaternary execution remains in Phases 10–11 and circuit execution in 12–13. Their inclusion in the journal milestone is unchanged. A prospective research idea recorded in the earlier novelty PDF is not implementation evidence.

Phase 5 preparatory design: guided and perturbative BP

Date: 26 September 2026. Status: proposed contracts; production execution has not started. Phase 4 remains active. Requirements: MOD-02, MOD-03, INT-01, INT-02, INT-03, INT-04.

1. Research boundary and source decisions

This design extends the canonical binary CSS engine. It introduces explicit changes to priors, constraints and attempt state while keeping the original observation immutable. It does not implement a second BP engine or substitute a native decoder for canonical policy execution.

Primary sources: BPGD, arXiv:2312.10950v2, Section 4.3 and Algorithm 1 (https://arxiv.org/html/2312.10950v2) and impulse decoding, arXiv:2606.18240v1, Section IV and Algorithm 1 (https://arxiv.org/html/2606.18240v1). The versioned source-status record is docs/evidence/phase5-source-status-20260926.json.

BPGD alternates SPA rounds and reliability-guided changes to channel priors; the reported finite magnitude is 25. Algorithm 1 assigns zero posterior bias to positive prior, whereas Equation 36 assigns it negative prior. Its last decimation is followed by return, without another explicit BP call. We propose an Algorithm-1 interpretation with declared zero and final-round rules. Continuing message state is an inference requiring native confirmation.

Impulse first tries BP, then independently shortens variables to one on nonconvergence and ranks converged candidates by Hamming weight. Its first-convergence alternative uses declared index order. Structural shortening below is our proposed exact-constraint realization; matching native numerical trajectories remains a separate question.

Neither source's native-code identity has been verified in this update. Syndrome/sign-flip source variants also remain unresolved. Their general contract below is ready for implementation planning, but a named source claim requires an equation, version and availability record first.

2. Objects and boundaries

Proposed object	Required contents and invariant
InterventionSpec	Source variant, trigger, eligible targets, operator, value, lifetime, reset, budget and ordering
WorkingProblemVNext	Original identity, transformed H and syndrome, ordered column/row maps, transformation and lift identities
FixationMap	Unique original indices, binary values, parent map, lifetime and release rule
BpSessionState	Graph/prior identities, messages, beliefs, histories, schedule cursor, dtype and committed-unit index
InterventionProposal	Before-state identity, target, proposed new state/view, reserved work and child lineage

Proposed object	Required contents and invariant
LiftMap	Total deterministic reconstruction to original coordinates and any algebraic certificate
CandidateValidation	Working validity, lift result, independent original-syndrome validity and rejection reason
RestartSpec	Preserved fields, reset fields, attempt cap, stream identity and stopping scope

Use a new public schema and versioned ledger grammar for these additions. Historical manifests and ledgers retain their meanings. Policy-visible views contain only the original observation, allowed graph/priors and committed decoder state. Hidden physical errors and logical outcomes remain exclusively in the independent evaluator. Selection never inspects logical truth.

The main engine currently initializes a new attempt; it has no admitted transformed WorkingProblem. A round/session API must make continuation an explicit operation rather than reconstructing messages from an output vector.

3. Finite bias and exact fixing

finite_prior_bias assigns a finite signed prior to a selected original index, marks it ineligible for another decimation, and leaves the graph unchanged. It does not promise that every subsequent hard decision equals that sign. hard_constraint fixes a bit algebraically, changes the working graph and provides a lift. Neither operator is an alias for the other. Clipping infinity to a numerical bound is an explicitly labeled approximation.

For BPGD, propose reliability abs(posterior_llr), eligibility among undecimated indices, stable original-index ties and positive prior for zero under the Algorithm-1 interpretation. Reliability ordering is equivalent to binary marginal confidence for finite LLRs. Reject NaN; declare infinity handling. Fixation persists through its declared search lifetime. Each BP round runs at most T whole admitted units and checks original validity at declared boundaries. A within-round early stop and an extra final BP round are separate adaptations, not implicit optimizations. Specify message continuation and recomputed belief/prior dependencies before admitting a source variant.

Hard fixing has eligible original indices, binary values and release/reset events. A release creates a child working view from the immutable original problem and remaining fixations; it does not edit a parent graph in place.

4. Exact shortening and lifting

The following is an independent GF(2) derivation of the proposed contract. Partition original columns into fixed F and retained R, with fixed values a. For original equation H c = s, define:

```
H_work = H[:, R]
s_work = s XOR (H[:, F] a)
lift(y)[F] = a
lift(y)[R] = y
```

Then H_work y == s_work if and only if H lift(y) == s, provided ordered maps and fixed values match the certificate. Independent original validation is still mandatory: it detects incorrect maps or corrupted lift records.

After shortening, an empty check with residual one is an immediate contradiction. An empty check with residual zero may be retained or removed only with an explicit row map. If all variables are fixed, construct and validate the unique lifted candidate without inventing a BP iteration. Dependent rows and inconsistent original syndromes must remain visible.

tools/validate_intervention_design.py enumerates all binary 2-by-3 matrices, syndromes, fixed subsets, fixed values and free candidates. It independently checks 16,384 shortening/lifting cases. This proves the enumerated finite fixtures, not arbitrary decoder behavior or a published performance claim.

5. Impulse attempt controller

Trigger on baseline nonconvergence/original-syndrome invalidity observable to the decoder. A valid recovery with wrong hidden logical action cannot trigger an intervention. Preserve the baseline diagnostic candidate even if rejected.

Each attempted index gets an immutable clone of the declared baseline prior and its own working view, fresh BP state and attempt identity. The proposed exact shortening path fixes that index to one. A source implementation using infinite priors needs an explicit numerical convention and separate fidelity assessment. Preserve every failed, contradictory and denied attempt.

Lift and original-validate each candidate before selection. Minimum-weight selection uses original-coordinate Hamming weight, followed by stable declared attempt/index order and local candidate order.
First-convergence selection returns the earliest eligible declared attempt, regardless of worker arrival; either await all earlier attempts or execute in that order. A parallel race mode would be a different, explicitly named adaptation.

6. Syndrome and sign perturbations

A syndrome flip creates s_work = s XOR d and retains s. A working-valid y does not become an eligible original recovery by relabeling its syndrome. Default handling lifts by coordinate identity and rejects it if original validation fails. If a supplied correction t satisfies H t = d, a certified lift c = y XOR t restores the original equation. A row mask alone is not this certificate. A generic solver that manufactures t adds separately declared work and belongs to an explicit policy, not silent validation.

A sign perturbation declares which object changes: channel prior, V2C, C2V or posterior; target selection, magnitude, sign/zero convention and trigger are mandatory. It also declares whether outgoing messages and cached beliefs are recomputed, retained or invalidated. Flipping a prior does not automatically flip stored messages. Until a primary source is resolved, these are research adaptation operators without a named source-faithful variant.

The retained design fixtures include a working-valid/original-invalid syndrome flip and the valid certified lift, plus an all-fixed contradiction. They also show that a finite signed prior can be outweighed by incoming messages.

7. Restarts, continuation and randomness

The default restart resets messages, beliefs, schedule cursor, residual cache and private memory histories. Preservation is allowed only for named fields with compatible graph, coordinate, prior, dtype and ownership identities. BPGD round continuation is a separate declared transition. Graph changes invalidate edge-indexed histories unless a checked mapping is provided.

Streams derive from root identity plus decoder, intervention, attempt and purpose namespaces. Physical sampling uses its independent namespace and cannot be consumed by ranking, branching or restarts. Replays record stream identity and realized draws. Worker count and scheduling do not change draws. When a failed run has a diagnostic candidate but no reliability field, a reliability-driven policy must fail clearly or use a prespecified alternative; it cannot invent a posterior. No-proposal is retained as its own terminal kind.

8. Atomicity, provenance and work

Reserve the declared unit before changing graph, priors, histories, ordering or random cursors. Validate the proposal, compute in private state and commit the child state plus events together. Budget denial leaves the parent intact. Numerical failure retains the last committed diagnostic state and records failed work according to its explicit accounting contract.

Retain links from original -> transformation -> working view -> attempt -> candidate -> lift -> original validation -> selection -> evaluation. Event payloads include source variant, target/value, parent state, map/certificate, reset/preservation rules, proposal and committed state digests, budget scope, denied work, actual work and terminal reason. Independent replay reconstructs semantics as well as hashes.

Count BP calls, C2V/V2C elements, belief assemblies, ranking scans, fixation actions, transformations, lifted coordinates and original checks separately. Aggregate actual leaf work once across restarts/attempts. A parent's subtotal is not an additional leaf contribution. Contradiction checks cost real work but no BP calls. Record setup, decode, lifting/validation, serialization and analysis boundaries independently; retain absent native counters as missing.

9. Evaluation and timing contract

Use matched physical samples with finite, prespecified attempt/work/shot/time caps and disjoint development/calibration/evaluation namespaces. Freeze coefficients and tie rules before outcomes. Include every admitted shot in all-shot failure LER, counting absent selected recovery as failure. Report valid-correct, valid-wrong-logical, invalid-correct-logical-projection, invalid-wrong-logical-projection and no-proposal separately. The invalid projection cells are basis-dependent diagnostics, not successful recovery.

For runtime, time both implementations from an already constructed decoder input to a returned raw candidate. Separate preparation and independent validation. Match update algorithm, numerical bounds, stopping, state, threads and work limits; counterbalance call order and disclose warmup. Report work distributions and outcomes alongside latency. Even then, a language-only causal claim needs implementation controls beyond a single Rust/Python pair. Do not tune settings on the runtime evaluation samples.

10. Gates and later reuse

The task map in 05-DESIGN-TASKS.md covers each Phase 5 requirement and success criterion. Source discovery, interface sketches and finite algebra validation can proceed now. Production Phase 5 execution requires completed Phase 4 and checked plans; named variants additionally require their source decisions. Source gaps remain visible rather than passing by an invented approximation.

Phase 6 may reuse working views, lifts and attempt lineage for search, but generic beam execution and voting remain its own work. Quaternary state and circuit observations require their mathematically compatible engines and later qualification. No CSS intervention result establishes those regimes.

Phase 5 design tasks and implementation handoff

Date: 26 September 2026. These are bounded preparatory tasks, not numbered execution plans. All six Phase 5 requirement IDs remain pending.

Completed design work in this update

- Reviewed Phase 4 dependencies and gaps against code/evidence.
- Defined intervention, working-view, session, lift and restart contracts.
- Recorded two BPGD source ambiguities and impulse realization boundaries.
- Established deterministic shortening and original-validation fixtures.
- Defined state rollback, work categories, selection and timing boundaries.

Next task map

Task	Responsibility and proposed files	Requirement	Dependencies
D1	Source decisions and availability records; data/evidence/phase5-source-decisions.json	MOD-02, MOD-03, INT-02, INT-03	Primary-source discovery; can proceed now
D2	Versioned working problem, fixation and lift contracts; canonical/problems.py, future transformations.py	INT-01, INT-02, MOD-03	D1 operator definitions; Phase 4 gate for execution
D3	Vector priors and BP round/session transition; canonical/kernel.py, future intervention_session.py	MOD-02, INT-03, INT-04	D2 and corrected memory-state ownership
D4	Source-qualified BPGD controller; future guided_decimation.py	MOD-02	D1, D3; explicit zero/final-round decisions
D5	Impulse attempt orchestration, lift and stable selection; future impulse.py	MOD-03	D1, D2, D3
D6	Freezing, source-qualified flips and restarts; future interventions.py	INT-01, INT-02, INT-03, INT-04	D1, D2, D3
D7	Public schema/ledger verifier, matched CSS fixtures and guide	All six	D4–D6; admitted finite pilot design

Acceptance checks to carry into execution plans

D1: Resolve paper versions/equations and zero/tie/final-round/numerical rules. Distinguish finite-bias BPGD from exact constraints. Find and pin native code, or retain an explicit unavailable/unverified status. Do not infer no implementation exists from a failed search. Source-defined syndrome/sign flips remain open until

their primary algorithms are selected.

D2: Preserve originals; demonstrate shortening equivalence, map bijection, all-fixed success/contradiction, dependent/empty checks and bad lift rejection. Exercise a working-valid/original-invalid candidate. The standalone finite algebra validator supplied now becomes an independent fixture oracle.

D3: Verify retained versus reset messages/histories, changed prior fields, graph/cache invalidation and private RNG cursor. Denied units leave all parent state unchanged; failures retain last committed diagnostics. The public memory dispatch correction must be incorporated before memory is composed.

D4: On constructed graphs, check selected reliability, deterministic ties, finite prior update, exclusion of decimated indices, T-unit round boundary, declared zero rule, last-round behavior and original stopping. Reject hidden truth in trigger and selection views. Preserve adapted variants as such.

D5: Check baseline-only success, failure trigger, independent attempts, fixed-one contradiction, actual attempt caps, lifting, minimum original weight, stable first-index alternative and worker-arrival invariance. Compare an eligible pinned native implementation only when available.

D6: Check fixation lifetime/release, syndrome certificate or original rejection, sign target/value and cache consequences, independent restart streams and reset/preservation compatibility. Count work from failed and denied paths honestly, without fabricating completed operations.

D7: Public manifests reject unsupported combinations; malformed/rehashed semantic changes fail verification. Enable/disable interventions on matched CSS development samples under finite caps, expose five outcome cells and candidate origins, and execute a bounded guide. Pilot labels remain adaptation/not_assessed; no curve agreement follows from fixture success.

Source and measurement decisions still open

- Confirm native BPGD message continuation and resolve Algorithm 1 versus Equation 36 at zero, plus the final round boundary.

- Pin impulse code and its infinite-prior convention; establish whether structural shortening is a source-equivalent realization or adaptation.

- Resolve exact syndrome/sign-flip algorithms; generic contracts are not enough to satisfy a source-defined requirement.

- Choose public schema/ledger versions and a typed compatibility matrix.

- Freeze pilot tolerances, attempt/work budgets and runtime protocol before sampling. Completed Phase 4 low-noise outcomes cannot choose these settings.

Dependency decision

Design does not require calibration to have a winner. Formal Phase 5 execution depends on the completed Phase 4 capability and evidence gates. The immediate coding continuation is the isolated Phase 4 ablation/trace work described in its handoff; D1 and interface design can advance alongside it.